The strict M_4 resultant-mean inequality

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Abstract

We prove both parameter branches of Mihoković–Raïssouli's M_4 sign conjecture. A mixed derivative reduces the assertion to positivity of one explicit function. Analytic estimates cover a neighborhood of zero and the entire infinite tail; a reproducible interval-arithmetic certificate covers the intervening compact interval. The finite computation encloses whole intervals, rather than sampling points.

Task 1: the original problem

For $a, b > 0$, define

$$M(a,b) = M_4(a,b) = \frac{b-a}{\sqrt{2}\,\operatorname{arsinh}((\log b - \log a)/\sqrt{2})} \quad (a \neq b), \qquad M(a,a) = a.$$

For $r \neq 0$, let $B_r(a,b) = ((a^r + b^r)/2)^{1/r}$, and put

$$\mathcal{R}(B_p, M, B_q)(a,b) = B_p\big(M(a, B_q(a,b)), M(B_q(a,b), b)\big).$$

The intended strict comparison is for distinct positive inputs; the diagonal has equality. This is Conjecture 4.10 of the preprint [1], and Conjecture 40, p. 22, of its complete published version.

Theorem 1. *If $p = \sqrt{21}$ or $p = -\sqrt{21}$, and $q = (3-p)/2$, then*

$$M(a,b) > \mathcal{R}(B_p, M, B_q)(a,b) \qquad (a, b > 0,\ a \neq b).$$

Thus both choices $q = (3 \pm \sqrt{21})/2$, $p = 3 - 2q$, in the conjecture are valid.

A mixed-derivative reduction

By symmetry, assume $a < b$, and set

$$H(a,b) = (b^q - a^q)M(a,b)^p, \qquad H(a,a) = 0.$$

We will prove $H_{ab} > 0$ for $a < b$. For any $a < c < b$, the fundamental theorem of calculus then gives

$$H(a,c) + H(c,b) - H(a,b) = \int_a^c \int_c^b H_{ab}(u,v)\,dv\,du > 0. \tag{1}$$

Take $c = B_q(a,b)$, so that $c^q - a^q = b^q - c^q = (b^q - a^q)/2$. If $q > 0$, (1) gives $M(a,b)^p < (M(a,c)^p + M(c,b)^p)/2$; here $p < 0$, giving the claimed strict comparison. If $q < 0$, the first inequality reverses on division by $b^q - a^q$; now $p > 0$, giving the same conclusion.

Here is an explicit test for the mixed derivative. For $t > 0$, write

$$v(t) = \log \frac{\sqrt{2}\sinh(t/2)}{\operatorname{arsinh}(t/\sqrt{2})}, \quad g(t) = \frac{1}{\sqrt{t^2+2}\,\operatorname{arsinh}(t/\sqrt{2})}, \quad s = p + q.$$

Then

$$v'(t) = \tfrac{1}{2}\coth(t/2) - g(t), \tag{2}$$

$$v''(t) = -\frac{1}{4\sinh^2(t/2)} + \frac{t}{t^2+2}g(t) + g(t)^2. \tag{3}$$

Define

$$T_p(t) = v''(t) + p\,v'(t)^2 + q\coth(qt/2)v'(t) - \frac{3}{4}. \tag{4}$$

With $t = \log(b/a)$, we have

$$H(a,b) = a^s h(t), \qquad h(t) = e^{st/2}\,2\sinh(qt/2)e^{pv(t)}.$$

Since $p + 2q = 3$, direct differentiation gives

$$h'' - sh' = phT_p, \qquad H_{ab}(a,b) = -\frac{a^s}{ab}\,ph(t)T_p(t).$$

Indeed, the constant in the first identity is $(q^2 - s^2)/4 = -3p/4$. The sign of $h(t)$ is the sign of q, and $pq < 0$. It remains to prove $T_p(t) > 0$ for $t > 0$, for both values of p.

A rigorous certificate for $T_p > 0$

All interval computations below use outward enclosures in Arb, through python-flint [2], at 160-bit midpoint precision. The accompanying `verify_inequality.py` recomputes the coefficients, partitions, and strict enclosures, and writes `exact-certificate.json`. The analytic estimates below supply the two infinite families of arguments outside the compact computation.

1. An analytic Taylor remainder at zero

The function v' is odd and analytic near zero. Its expansion begins

$$v'(t) = \frac{t}{4} - \frac{t^3}{16} + \frac{17t^5}{672} + O(t^7).$$

Consequently T_p is even and analytic, and, using $p^2 = 21$,

$$T_p(t) = \frac{13}{480}t^4 + O(t^6). \tag{5}$$

For explicit verification, the coefficients of t^2 and t^4 are

$$\frac{-15 + 3p + 2q^2}{48} = 0, \qquad \frac{255 - 15q^2 - q^4 - 45p}{1440} = \frac{13}{480}.$$

The constant coefficient is zero as well.

We need a uniform, proved bound on all omitted coefficients. Define

$$\rho(z) = \frac{\operatorname{arsinh}(z/\sqrt{2})}{z/\sqrt{2}}, \qquad U(z) = \frac{1}{\sqrt{1+z^2/2}\,\rho(z)}, \qquad D(w) = w\coth w.$$

The removable values at zero are one. For $|z| \le 1$, the binomial series gives

$$|\rho(z) - 1| \le \frac{1}{3}, \qquad |(1+z^2/2)^{-1/2} - 1| \le \sqrt{2} - 1 < \frac{1}{2}.$$

Thus $|U - 1| < 5/4$. Also, on $|z| \le 1$,

$$|D(z/2) - 1| \le \frac{(\cosh(1/2) - 1) + (\sinh(1/2)/(1/2) - 1)}{1 - (\sinh(1/2)/(1/2) - 1)} < \frac{1}{4}.$$

For example, the two numerator terms are less than $1/7$ and $1/20$. The denominators in these formulas have no zeros in the specified discs. Since $v'(z) = (D(z/2) - U(z))/z$, the maximum principle gives

$$|v'(z)/z| < \frac{3}{2} \quad (|z| \le 1).$$

Cauchy's derivative estimate therefore implies, on $|z| \le 1/2$,

$$|v'(z)| < \frac{3}{4}, \qquad |v''(z)| < 3.$$

Furthermore, $|q| < 4$, and for $|w| \le 1$,

$$|D(w)| \le \frac{\cosh 1}{1 - (\sinh 1 - 1)} < 2.$$

Here $\cosh 1 < 8/5$ and $\sinh 1 - 1 < 1/5$. Writing $q\coth(qz/2)v'(z) = 2D(qz/2)v'(z)/z$, we obtain

$$|T_p(z)| < 3 + 5(3/4)^2 + 6 + 3/4 < 16 \qquad (|z| \le 1/2).$$

These estimates also establish analyticity throughout that disc. If $T_p(t) = \sum a_k t^k$, Cauchy's bound is $|a_k| \le 16\,2^k$. Parity and (5) now give, for $0 < t \le 1/4$,

$$\left| \frac{T_p(t) - \sum_{k=2}^{20} a_{2k} t^{2k}}{t^4} \right| \le \frac{16\,2^{42} t^{38}}{1 - 2t} \le 2^{-29}. \tag{6}$$

The verifier computes the coefficients through degree 40 by formal ball-series arithmetic on the displayed formulas. Horner evaluation of $\sum_{k=2}^{20} a_{2k} y^{k-2}$ on the single interval $0 \le y \le 1/16$, followed by subtraction of 2^{-29}, gives a lower bound greater than $1/40$, for each branch. It follows that

$$T_p(t) > \frac{t^4}{40} > 0 \qquad (0 < t \le 1/4).$$

Formal-series division by t removes only the exact zero terms justified by parity and the displayed removable values.

2. The compact interval

Start with the exact rational interval $[1/4, 20]$. For each interval $[l, u]$, evaluate (2)–(4) on the Arb ball centered at $(l+u)/2$ with radius $(u-l)/2$. Accept the interval only when the entire resulting enclosure is strictly positive; otherwise bisect it at its exact rational midpoint and repeat. The computation terminates:

p	accepted subintervals	certified lower bound on every leaf
$-\sqrt{21}$	186338	10^{-12}
$\sqrt{21}$	173452	10^{-12}

The code checks that the exact sum of accepted lengths is $79/4$. The binary subdivision preserves a cover with disjoint interiors; thus this is a certificate for every point of $[1/4, 20]$. The JSON certificate records partition hashes and depth counts. No unverified decimal approximation of p, q, or an endpoint is used.

3. The whole tail $t \geq 20$

Let $r = \sqrt{21}$, $k = (r-3)/2$, and $\epsilon = (e^t - 1)^{-1}$. We have

$$\frac{3}{4} < k < 1, \quad 4 < r < 5, \quad v' = \frac{1}{2} - g + \epsilon, \quad v'' = g^2 + \frac{t}{t^2+2} g - \epsilon(1+\epsilon).$$

For $t \geq 20$,

$$\frac{1}{2t\log(2t)} < g < \frac{1}{t} \leq \frac{1}{20}, \qquad 0 < \epsilon < 2e^{-t}.$$

The first lower bound follows from $\sqrt{t^2+2} < 2t$ and $\operatorname{arsinh}(t/\sqrt{2}) < \log(2t)$. In particular $v' > 0$.

For $p = -r$, $q = (3+r)/2$, use $\coth(qt/2) > 1$ in (4) and expand. Dropping positive terms gives

$$\begin{aligned} T_{-r} &\geq kg - (r-1)g^2 - (k+1)\epsilon - (r+1)\epsilon^2 \\ &> \frac{1}{2} g - 2\epsilon - 6\epsilon^2 \\ &> \frac{1}{4t\log(2t)} - 4e^{-t} - 24e^{-2t} \\ &\geq \frac{1}{4t^2} - 28e^{-t} > 0. \end{aligned}$$

Here $k - (r-1)g > 3/4 - 4/20 > 1/2$, and $\log(2t) < t$. For the final strict inequality, $t^2 e^{-t}$ decreases for $t \geq 20$, while

$$e^{20} > \frac{20^8}{8!} > 44800 = 112 \cdot 20^2.$$

For $p = r$, $q = -k$, we instead have $q\coth(qt/2) > k$ and $v' > 9/20$. Consequently

$$T_r > -\epsilon(1+\epsilon) + 4(9/20)^2 + \frac{3}{4}(9/20) - \frac{3}{4} = -\epsilon(1+\epsilon) + \frac{159}{400} > 0.$$

Indeed $\epsilon < 2e^{-20} < 1/100$, so its negative contribution is less than $101/10000$. This completes the proof that $T_p > 0$ for every $t > 0$. The mixed-derivative reduction and (1) prove Theorem 1. $\square$

Scope of the proof and audit. This is a computer-assisted proof: its compact and near-zero polynomial enclosures are substantive proof components. The script is independently rerunnable and supplies rigorous enclosures, not merely high-precision samples. The authoring agent checked the algebra, analytic tails, source scope, and executed the certificate. Independent peer review and proof-assistant formalization are not claimed.

Proposed follow-up questions

The following four questions are unproved directions, not additional solves.

Task 2. Find an elementary analytic proof of $T_{\pm\sqrt{21}}(t) > 0$ that removes the interval computation.

Task 3. Determine the complete region of real parameter pairs (p, q) for which the inequality $M_4 > \mathcal{R}(B_p, M_4, B_q)$ holds for every distinct positive pair.

Task 4. Prove or disprove the analogous M_2 inequality in Conjecture 37 of the published source, for both prescribed power-mean pairs.

Task 5. For each branch here, determine the largest constant $C_p > 0$ such that

$$\frac{M_4(a,b) - \mathcal{R}(B_p, M_4, B_q)(a,b)}{M_4(a,b)} \geq C_p \left(\frac{|\log(b/a)|}{1 + |\log(b/a)|} \right)^6$$

for every distinct positive pair, and determine whether the infimum is approached on the diagonal.

References

[1] L. Mihoković and M. Raïssouli, *Results on comparison and sub/super-stabilizability of some new means*, Applicable Analysis and Discrete Mathematics **20** (2026), 1–23, doi:10.2298/AADM240618001M. Complete published text: journal PDF. Preprint: arXiv:2405.13618v2.

[2] The python-flint developers, *Arb real numbers and rigorous ball arithmetic*, official documentation. The accompanying certificate records the exact installed version.